

Informatics II, Spring 2026, Solution 5

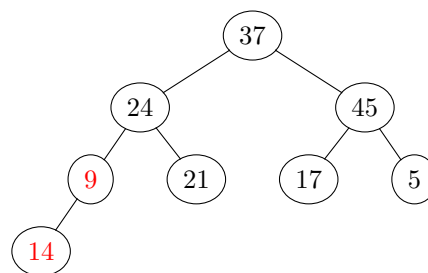
Task 1: Heapsort Algorithm

Making max-heap
Step 1

Input array

37	24	45	9	21	17	5	14
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Binary tree

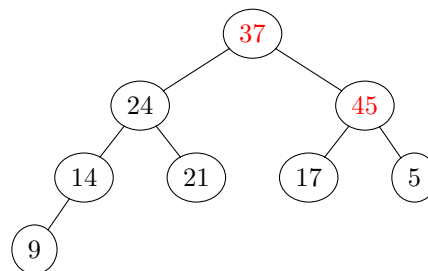


Step 2

Input array

37	24	45	14	21	17	5	9
----	----	----	----	----	----	---	---

Binary tree

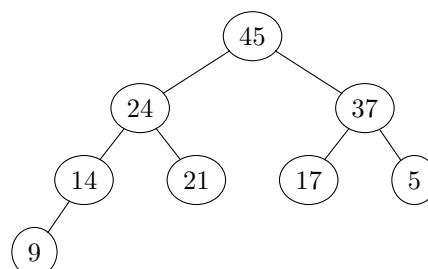


Max-heap

Input array

45	24	37	14	21	17	5	9
----	----	----	----	----	----	---	---

Max-heap

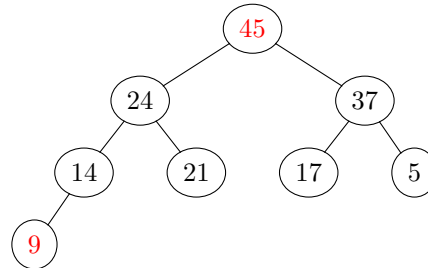


Heapsort
 Iteration 1
 Exchange last and first(root) element

Input array

45	24	37	14	21	17	5	9
----	----	----	----	----	----	---	---

Max-heap

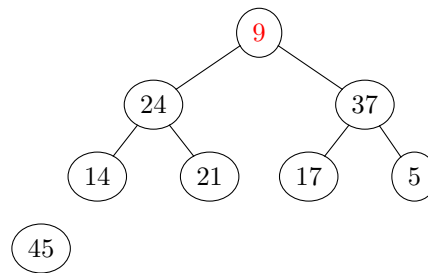


Reorder the tree to make max-heap again

Input array

9	24	37	14	21	17	5	45
---	----	----	----	----	----	---	----

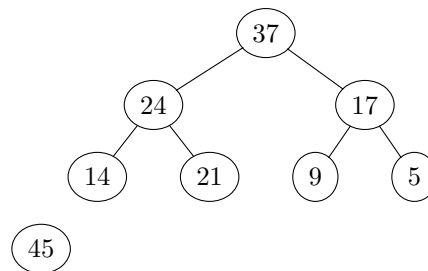
Max-heap



Input array

37	24	17	14	21	9	5	45
----	----	----	----	----	---	---	----

Max-heap

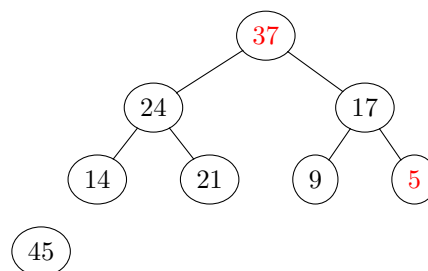


Iteration 2
 Exchange last and first(root) element

Input array

37	24	17	14	21	9	5	45
----	----	----	----	----	---	---	----

Max-heap

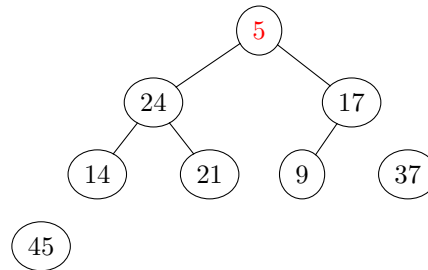


Reorder the tree to make max-heap again

Input array

5	24	17	14	21	9	37	45
---	----	----	----	----	---	----	----

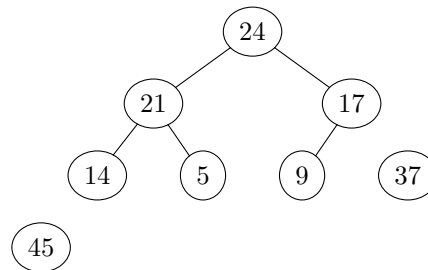
Max-heap



Input array

24	21	17	14	5	9	37	45
----	----	----	----	---	---	----	----

Max-heap



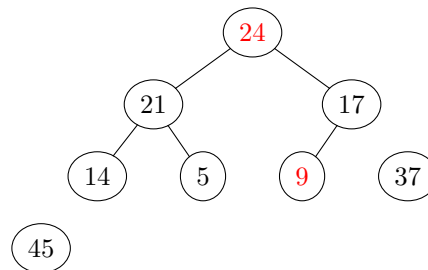
Iteration 3

Exchange last and first(root) element

Input array

24	21	17	14	5	9	37	45
----	----	----	----	---	---	----	----

Max-heap

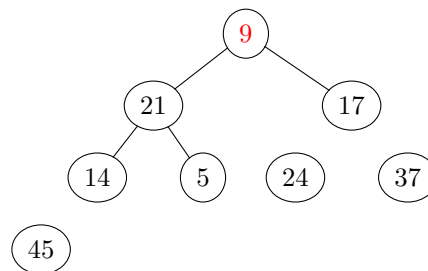


Reorder the tree to make max-heap again

Input array

9	21	17	14	5	24	37	45
---	----	----	----	---	----	----	----

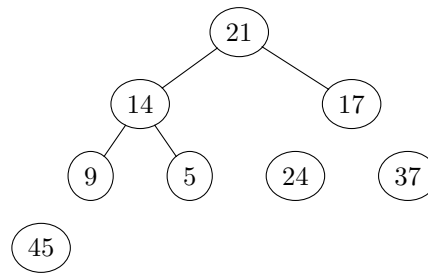
Max-heap



Input array

21	14	17	9	5	24	37	45
----	----	----	---	---	----	----	----

Max-heap

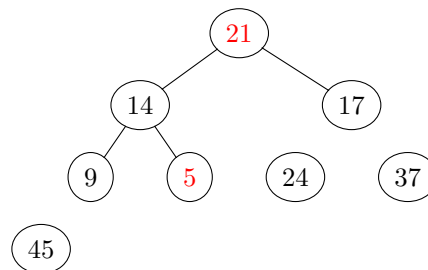


Iteration 4
Exchange last and first(root) element

Input array

21	14	17	9	5	24	37	45
----	----	----	---	---	----	----	----

Max-heap

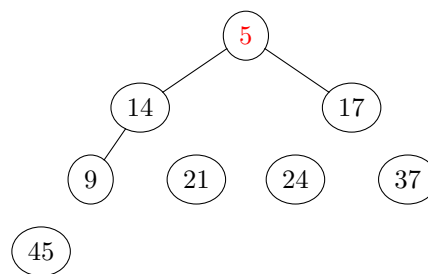


Reorder the tree to make max-heap again

Input array

5	14	17	9	21	24	37	45
---	----	----	---	----	----	----	----

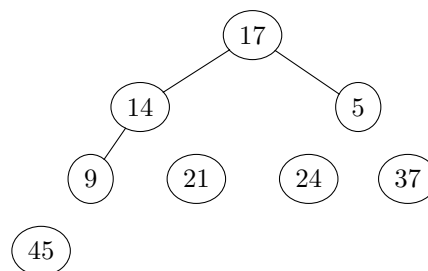
Max-heap



Input array

17	14	5	9	21	24	37	45
----	----	---	---	----	----	----	----

Max-heap

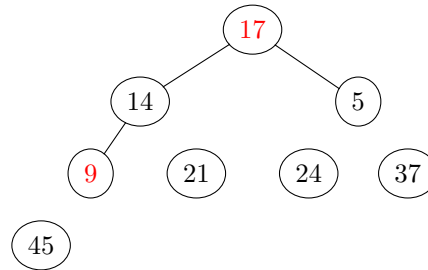


Iteration 5
Exchange last and first(root) element

Input array

17	14	5	9	21	24	37	45
----	----	---	---	----	----	----	----

Max-heap

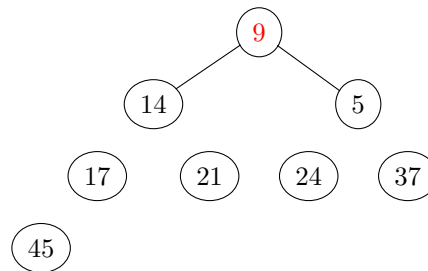


Reorder the tree to make max-heap again

Input array

9	14	5	17	21	24	37	45
---	----	---	----	----	----	----	----

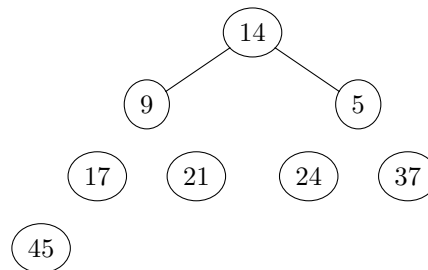
Max-heap



Input array

14	9	5	17	21	24	37	45
----	---	---	----	----	----	----	----

Max-heap

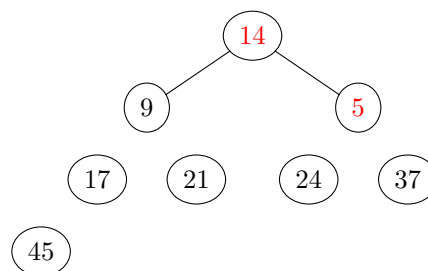


Iteration 6
Exchange last and first(root) element

Input array

14	9	5	17	21	24	37	45
----	---	---	----	----	----	----	----

Max-heap

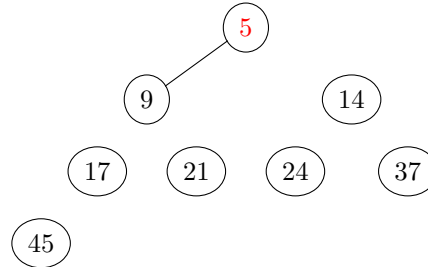


Reorder the tree to make max-heap again

Input array

5	9	14	17	21	24	37	45
---	---	----	----	----	----	----	----

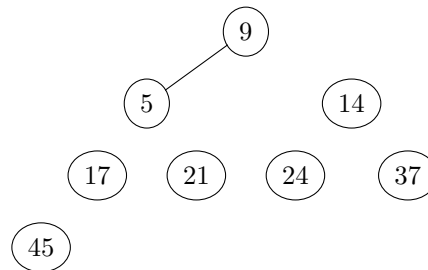
Max-heap



Input array

9	5	14	17	21	24	37	45
---	---	----	----	----	----	----	----

Max-heap



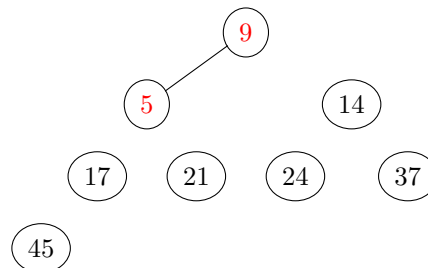
Iteration 7

Exchange last and first(root) element

Input array

9	5	14	17	21	24	37	45
---	---	----	----	----	----	----	----

Max-heap

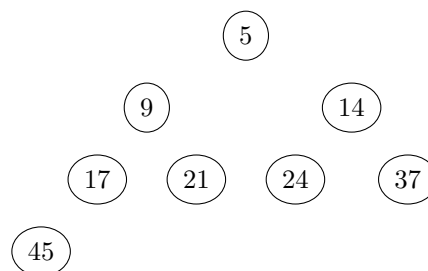


No need for reorder

Input array

5	9	14	17	21	24	37	45
---	---	----	----	----	----	----	----

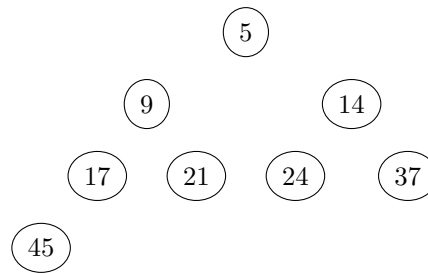
Max-heap



Sorted array

5	9	14	17	21	24	37	45
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Max-heap



Task 2: Heap and Heapsort

See code in the file task2.c

Task 3: Quicksort

See code in the file task3.c